

1. Problem Statement

Printed circuit boards (PCBs) are foundational components in virtually all electronic devices, and manufacturing defects such as open circuits, short circuits, missing holes, mouse bites, spurs, and spurious copper deposits can render entire assemblies non-functional or introduce latent reliability failures. Current inspection methods in many facilities still rely on manual visual review, which is slow, error-prone, and difficult to scale. This project aims to build an end-to-end computer vision pipeline that automatically detects, localizes, and classifies PCB defects from high-resolution board images. By combining CNN-based feature extraction with real-time object detection (YOLOv8), the system will provide fast, accurate defect identification suitable for integration into automated optical inspection (AOI) workflows; ultimately improving manufacturing quality assurance and enabling proactive detection of counterfeit or tampered hardware in supply chain security contexts.

2. Dataset & Data Collection Plan

Primary Dataset:

- PCB Defect Dataset (Kaggle) - <https://www.kaggle.com/datasets/akhatova/pcb-defects>

Supplementary Datasets:

- DeepPCB (GitHub) - <https://github.com/tangsanli5201/DeepPCB>
- PCB-Components (Roboflow) - <https://universe.roboflow.com/nchowdarym/pcb-components-djhuq>

The primary Kaggle dataset will serve as the core training and evaluation set. DeepPCB provides paired defective/template images useful for template-matching comparison baselines with trained models. The Roboflow PCB-Components dataset enables a secondary component-identification task, allowing the pipeline to both locate components and detect defects within them.

3. Preliminary Data Profile

Dataset	Images	Classes	Format
PCB Defects (Kaggle)	1,386	6 defect types	Pascal VOC (XML)
DeepPCB (GitHub)	1,500 pairs	6 defect types	Custom TXT (bbox)
(Roboflow) PCB-Components	401	16 component types	COCO JSON / YOLO TXT

Defect classes (primary dataset): open circuit, short, mouse bite, spur, spurious copper, missing hole. Each image contains 3–5 defects of the same category. The dataset includes both aligned images and rotation-augmented variants to simulate real-world placement variability.

Combined totals: ~3,287 images across 22 unique classes (6 defect types + 16 component types). All

datasets will be normalized to YOLO TXT format for unified training.

4. Primary CV Techniques

- Data Augmentation & Preprocessing - Rotation, flipping, contrast adjustment, and other adjustments to address class imbalance and improve generalization across varying board orientations and lighting conditions.
- Image Classification (ResNet) - Transfer learning with pre-trained CNN backbone (ResNet) for defect-type classification on cropped regions of interest, providing a secondary validation stage after detection.
- Object Detection (YOLOv8) - Real-time defect localization and classification using a CNN backbone with anchor-free detection heads. Will serve as the primary detection model and performance baseline.
- Template Matching & Image Differencing - Leveraging DeepPCB's paired template/defective images to compute pixel-level difference maps as a classical CV baseline for comparison against the deep learning detection models.

Please note that Claude Opus 4.6 was used to generate this overview of the final project but research and identification of the dataset and project purpose was done by me personally.